Semtech (SMTC)

Production artifact snapshot: G:/My Drive/1amInvesting/Artifacts/SMTC_Semtech_DeepDive.pdf

CURRENT EDITION - 3 October 2026 | Framework 8.0

Full investment research and current probability-weighted operating model. The October 3 update incorporates the Tower expert capacity signal and issuer calibration without changing the saved forecasts, ceilings or Base scores. Financial assumptions, source evidence and forward estimates are distinguished throughout. Production snapshot as of 2026-10-03: G:/My Drive/1amInvesting/Artifacts/SMTC_Semtech_DeepDive.pdf.

Q4 2027 ceiling

Current model item	Value
Q4 2027 ceiling	205.1131 USD/share
Valuation window	Q4 2027 to Q3 2028
Artifact analytical Base	78/100
Protected portfolio Excel Base	78/100

A small optical-component pricing contribution is limited by the existing 72% data-center gross-margin cap. Copper and other businesses receive no direct policy benefit.

The sanctioned valuation remains **20x forward-four-quarter operating profit**, with the existing accounting basis, ownership, shares and currency conversion. The active ceiling next rolls on January 1, 2027.

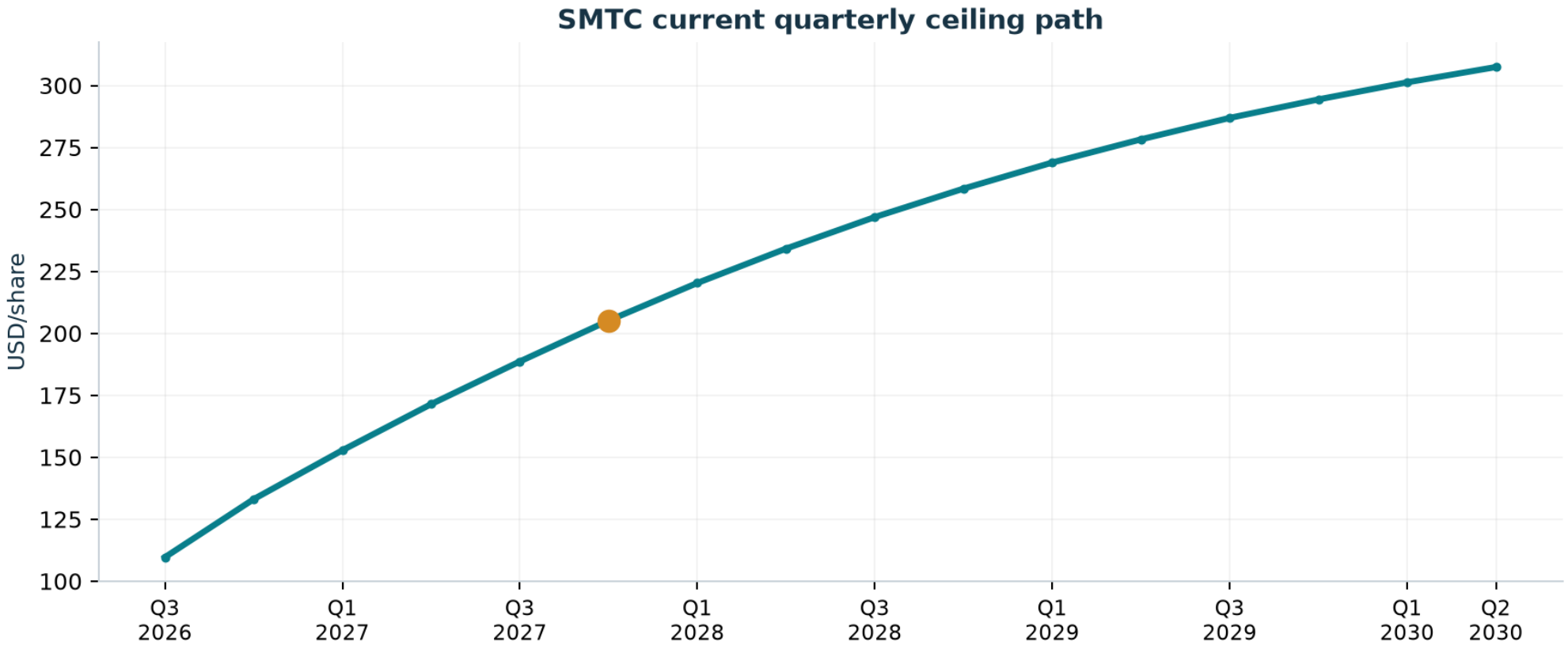


Figure 1. Current quarterly ceilings, USD/share. Gold marks the active Q4 2027 horizon. The 40% policy contribution is included once in operating earnings.

Ceiling quarter	Current ceiling USD/share
Q3 2026	109.6593
Q4 2026	133.0743
Q1 2027	152.8108
Q2 2027	171.4197
Q3 2027	188.6449
Q4 2027	205.1131
Q1 2028	220.3035
Q2 2028	234.1612
Q3 2028	246.8823
Q4 2028	258.3624
Q1 2029	268.8895
Q2 2029	278.2460
Q3 2029	286.9007
Q4 2029	294.3595
Q1 2030	301.2580
Q2 2030	307.4247

Analytical Base and scored conviction

Category	Score	Evidence and limit
Core Thesis	16/20	FiberEdge 800G share moved from roughly 18% two years ago to **well over 50%**, on an industry unit base that went from about 20 million transceivers a year to roughly 90 million. 1.6T FiberEdge share is targeted **above 50% by fiscal year end**. The company is **designing to every module provider in its target markets, several on a sole-source basis**. CopperEdge linear equalizers are the **de facto industry standard** with design-in at all bandwidths up to 3.2T. Held below the chokepoint band because Rule 9 fails: merchant competitors exist and no physics or patent gate gives sole supply
Cycle Stage	10/10	Tenth consecutive quarter of net sales growth, record revenue above the high end of outlook, +33% YoY, guided +54% YoY with data centre +160%. FY2027 backlog **all booked** against target and FY2028 **over 70% booked**
TAM Expansion	9/10	Content per transceiver guided from high single-digit dollars to **80 to 90 dollars** as the industry moves 800G to 3.2T, a roughly tenfold content step. Portfolio now spans gain chips, high-power CW lasers, SOAs and photodiodes for scale-up, scale-out and scale-across. Held below 10 because it remains component content inside someone else's system
Catalysts	5/5	Investor event 15 October with a multi-year financial framework; divestiture close in FQ4 FY2027; CW laser revenue from H1 FY2028; ACC volume deployment from FQ4; 3.2T design window opening within 12 months
Macro	9/10	DB1 photonics and AI interconnect. Held below 10 because the company depends on hyperscaler capex it does not control
Alpha	2/5	Well-covered Nasdaq name with broad sell-side attention. No local-language edge and no coverage gap
Risks	-4	Four classes, one point each. See section 6
Total	47/60 = 78/100	High Conviction; price-independent analytical thesis score

Artifact analytical Base: 78/100. Protected portfolio Excel Base: 78/100. The portfolio value is a separate retained decision; it is not a scoring-table input.

Qualified optical-component content remains attractive, but competition and outsourced supply persist. The policy contribution is constrained by the existing 72% data-center gross-margin cap. No new dated customer award, capacity commitment or defensible additional monopoly credit is established.

Base measures thesis quality. The 40% policy probability weights only the incremental operating economics, once; neither Base nor the policy weight is applied again to the final ceiling. Scores use integer category points on the /60 framework lattice.

October 3 capacity evidence and model disposition

Decision: retain the complete operating path and valuation. The precise Tower allocation hypothesis receives 30%, while the claimed doubling is fully overlapped with the existing revenue ramp. The evidence increases the auditability of the supply assumptions; it does not establish incremental sales above the saved forecast. The active Q4 2027 ceiling remains **USD 205.1131/share**, on **20x** Q4 2027-Q3 2028 adjusted operating profit and **101.8 million** ceiling-quarter diluted shares. Analytical and portfolio Base remain **78/100**.

The October 2 P Equity Research / tuolaji expert call is a single publisher/expert chain, with an unidentified expert and AI translation. Its specific capacity, customer funding and die-economics claims are useful hypotheses, not issuer-reported commitments. The three 30% weights below apply to their narrow stated outcomes; they are not probabilities that Semtech's whole business succeeds, are not independent corroborations and are not applied to the final ceiling. The retained FCC pricing branch remains separately weighted at 40% and is already included once in the saved path.

Issuer evidence versus the precise expert claim

Semtech reported USD 100 million of FQ2 FY2027 data-center revenue and guided roughly USD 145 million for FQ3. Management's August 25 call described room for 50-100% capacity growth, prepayment/capex discussions and additional backend testers. It did not publish an exact 8,000-wafer Tower allocation or the expert's funding amount. HieFo's separate 3-4x fab expansion is not a Tower SiGe multiplier. Capacity must convert through qualified good die, packaging/test, customer demand and revenue recognition.

The expert describes a 3,500-4,000 monthly starting allocation and approximately 8,000 by 2027, but also describes adding 5,000 (which would imply 8,500-9,000). PDF page 17 uses both mid-2027 and end-2027. The conditional bridge uses the supplied 8,000 endpoint and a late-2027 interpretation, while retaining these inconsistencies explicitly.

Semtech's demand has historically been quite stable, averaging about 3,500 wafers per month. But around August 2026, the company raised new requirements, wanting 8-inch monthly capacity to increase by 5,000 wafers by mid-2027, raising the current monthly average of 3,500–4,000 wafers to 8,000 wafers per month.

Q23. What specific plans does the customer have for 2027 capacity demand for

5th-generation SiGe TIA chips used in 1.6T optical modules? What are the underlying drivers and chip-usage logic?

The customer hopes to ramp related capacity up to 8,000 wafers per month by the end of 2027. This demand is mainly driven by 1.6T optical modules, centered on 5th-generation SiGe TIA. Specifically, 1.6T optical modules use an 8×200G architecture, and the TIA used follows a "one-drives-four" configuration, so each 1.6T module needs two TIAs and two Drivers — four chips in total. Compared with the previous solution, chip usage has tripled, so total demand has grown twelvefold. Based on this, the customer's request to double capacity is reasonable. This increase will directly drive revenue growth.

Q24. Regarding the capacity demand brought by 1.6T optical modules, what are the related expansion plans, investment scale, cost structure, and the cooperation model reached with the customer?

To meet demand, the plan is to add 5,000 wafers per month of SiGe capacity. The total budget for the expansion project is a little over $90 million, of which the customer has been asked to contribute more than $60 million in the form of Capex, and the customer has agreed. This customer investment does not need to be repaid and will be used to amortize depreciation on the new capacity. Regarding the cooperation agreement, although expansion funding is mainly provided by the customer, the agreement stipulates that if the customer does not use

Source exhibit C1. User-supplied P Equity Research / tuolaji expert call, October 2, 2026, PDF page 17. Anonymous expert claims and AI translation; precise allocation, timing and funding are unconfirmed. Full source page retained.

Calibrated optical pool and the required mix / realization hurdle

The public DC anchor combines optical and copper products. Applying the retained 60% optical proxy produces a USD 60 million anchor. Calibrating this pool to 3,750 wafers/month gives USD 5,333.33 of realized optical revenue per wafer. At 8,000 wafers/month and unchanged realization, quarterly optical revenue would be USD 128 million. The saved Q4 2027 optical proxy is USD 171.2565 million, requiring **1.33794141x** the calibrated realization. This is a required mix/content/utilization hurdle, not a newly assumed universal ASP increase.

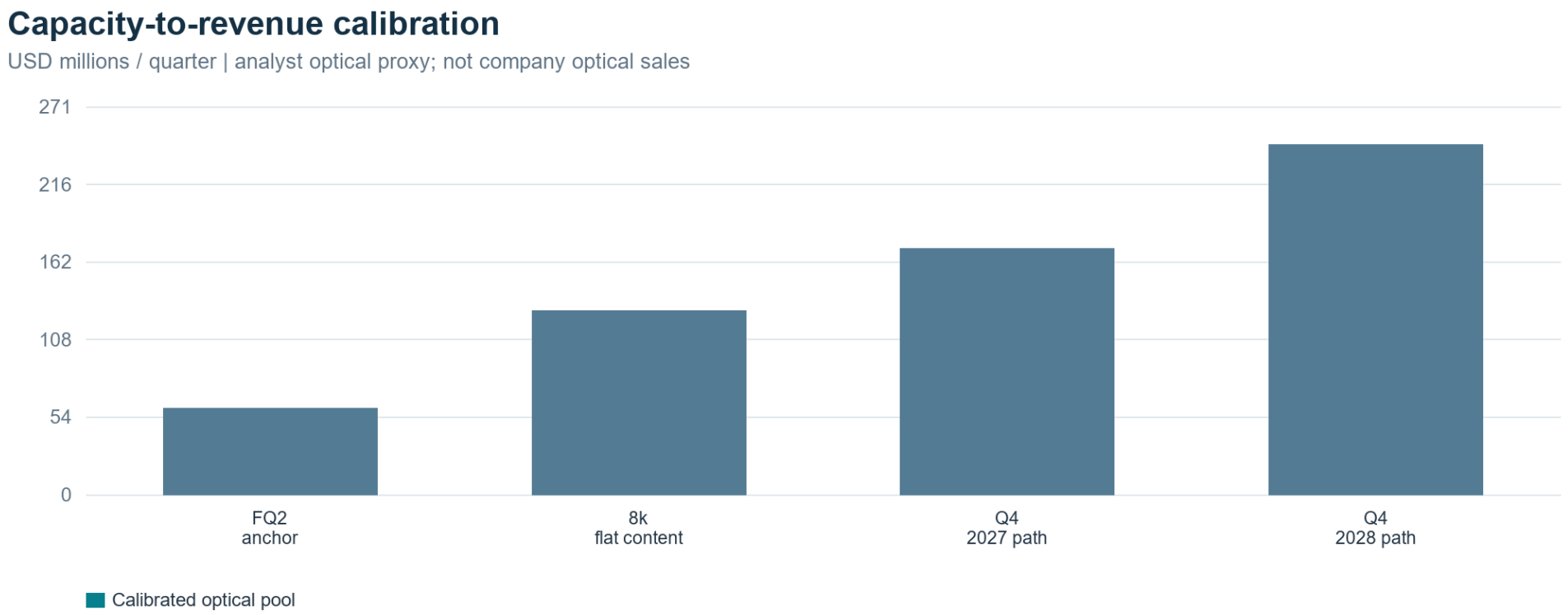


Figure C2. Calibration of the attributed wafer endpoint against the retained optical revenue proxy. The Q4 2027 and Q4 2028 bars are existing forecast requirements; the flat-content endpoint is an arithmetic reference, not an imposed revenue cap. Source: frozen model and attributed expert inputs, October 3, 2026.

Bridge item	Value	Definition / status
Reported FQ2 FY2027 DC anchor	100.0000	USD M / quarter; includes copper and photonics
Optical share proxy	60.00%	Analyst allocation; not company-disclosed
Calibrated optical anchor	60.0000	USD M / quarter
Expert starting allocation midpoint	3,750	Wafers/month; midpoint of attributed 3,500-4,000
Calibrated optical revenue per wafer	5,333.3333	USD; 60M / (3,750 x 3); realization proxy, not ASP
Expert conditional endpoint	8,000	Wafers/month; late-2027 interpretation, 30% hypothesis
Endpoint at unchanged realization	128.0000	USD M / quarter; 8,000 x 3 x calibrated value
Saved Q4 2027 DC path	285.4275	USD M / quarter; frozen forecast
Saved Q4 2027 optical proxy	171.2565	USD M / quarter; 60% of saved DC path
Required Q4 2027 realization factor	1.33794141x	171.2565 / 128; mix/realization hurdle, not verified ASP
Saved Q4 2028 DC path	406.2150	USD M / quarter; frozen forecast
Saved Q4 2028 optical proxy	243.7290	USD M / quarter; 60% of saved DC path
Q4 2028 factor if capacity held at 8,000	1.90413281x	Later hurdle only; no 8,000-wafer cap imposed on forecast

If monthly capacity stayed at 8,000, the saved Q4 2028 optical proxy would require **1.90413281x** the anchor realization. The report does not impose that capacity ceiling on later forecasts. Future wafer expansion, product content, realized mix and backend conversion remain evidence to test. None is silently added again as a separate profit branch.

Raw expert wafer x die x chip arithmetic produces **USD 304.5-432 million per quarter today**, already 3.05-4.32x total reported DC revenue, and **USD 696-864 million** at 8,000 wafers/month. These are gross-production-value calculations with unmatched scope and realization, not revenue estimates. USD 2,400 per wafer divided by 5,000-6,000 dies gives USD 0.40-0.48 raw wafer cost per die; it excludes yield losses, packaging/test and other costs. It cannot establish a corporate gross margin above the retained 72% DC cap. Quad and octal architectures coexist, so four chips per module is not a universal bill of materials.

Explicit probability and overlap ledger

PE-TSEM-20261002-SMTC-CAPACITY

Field	Disposition
Precisely weighted hypothesis	Approximately 8,000 monthly Tower SiGe wafers available to Semtech by late 2027
Probability	30%
Probability rationale	One credible publisher/expert chain; unidentified expert, AI translation and conflicting mid/end-year timing. Primary evidence supports capacity investment generally, not this precise amount.
Existing reference	100M quarterly DC anchor, rising to 285.4275M Q4 2027; model already assumes 50-100% capacity expansion and richer content.
Conditional interpretation	8,000 wafers/month with 1.33794140625x calibrated optical revenue per wafer supports the existing Q4 2027 optical pool. This factor is the required mix/realization hurdle, not a separately forecast or verified ASP increase.
Overlap / exclusion	100% of the claimed volume increase is absorbed by the existing shipment/content path; do not add 30% of the whole capacity endpoint.
Incremental revenue / GP / opex / OP	0 / 0 / 0 / 0 USD million in every modeled quarter
Evidence trigger	Increase incremental economics only if allocated/accepted wafer output, sellable die mix and customer revenue support output beyond this existing path. Reduce timing if backend capacity or qualification slips.

PE-TSEM-20261002-SMTC-FUNDING

Field	Disposition
Precisely weighted hypothesis	Semtech contributes more than USD60M to a more than USD90M Tower expansion
Probability	30%
Probability rationale	Specific single-source amount; issuer corroborates prepayment/capex route but not this amount or accounting.
Existing reference	Capacity investment and front/back-end constraints already described; future product GM includes production cost.
Conditional interpretation	Funding is a cash use, prepayment or capitalized right depending contract; it is not immediate revenue or profit. No separate project-life or repayment terms established.
Overlap / exclusion	No second profit/capex-derived revenue layer. Retain 72% DC GM cap and original costs; investigate recoverability/amortization when contract detail is available.
Incremental revenue / GP / opex / OP	0 / 0 / 0 / 0 USD million in every modeled quarter
Evidence trigger	Company disclosure of contract life, ownership, prepayment recoverability and expense recognition.

PE-TSEM-20261002-SMTC-DIE-ECONOMICS

Field	Disposition
Precisely weighted hypothesis	Selected 200G TIA chip ASP is around USD5.8-6 and raw wafer cost is about USD2400
Probability	30%
Probability rationale	Specific expert channel check; product and volume denominator uncertain.
Existing reference	64.8% modeled anchor DC GM; path capped at 72%.
Conditional interpretation	Literal current wafer x die x chip arithmetic yields USD304.5-432M quarterly, versus only USD100M reported DC revenue including copper. Calibrated proxy uses USD60M optical pool, not raw gross die count.
Overlap / exclusion	Cost per raw die excludes scrap, packaging/test, product mix, inventory and selling/engineering costs; it does not establish a company margin above the saved cap. Four chips/module is not universal because quad and octal architectures coexist.
Incremental revenue / GP / opex / OP	0 / 0 / 0 / 0 USD million in every modeled quarter
Evidence trigger	Matched finished-product shipments, good-die yield, exact die mix, packaging/test cost and realized ASP.

Funding is a cash-use hypothesis: the expert describes more than USD 60 million from Semtech toward a more than USD 90 million expansion. Public filings do not establish this specific contract, recoverability, ownership, life or accounting. It is not immediate profit, and no grant-funded second revenue layer is added. The existing adjusted-OP basis retains normal depreciation; capex, prepayments and depreciation are different items.

New capacity evidence effect	Active quarter Q4 2027	Forward window Q4 2027-Q3 2028	Model disposition
Revenue / GP / opex / OP	0 / 0 / 0 / 0	0 / 0 / 0 / 0	Every frozen forecast row unchanged
Ceiling / Base	205.1131 / 78	20x adjusted OP; 101.8M shares	Retained

Policy breadcrumb and economic hypothesis

Serenity's October 1 post attributes Washington meetings to Morgan Stanley and reports a possible 3.2T-focused restriction, with a potential 65% U.S.-value route for Chinese modules. The source expects laser pricing to matter more than broad module-share redistribution. Treat this as an attributed policy scenario; the original MS note and final FCC language establishing these exact terms were not recovered. The separate September 25 Senate proposal concerns sensitive federal systems and a five-year transition. It is not confirmation of these FCC terms.

Serenity policy post | Foundry follow-up | Separate Senate proposal

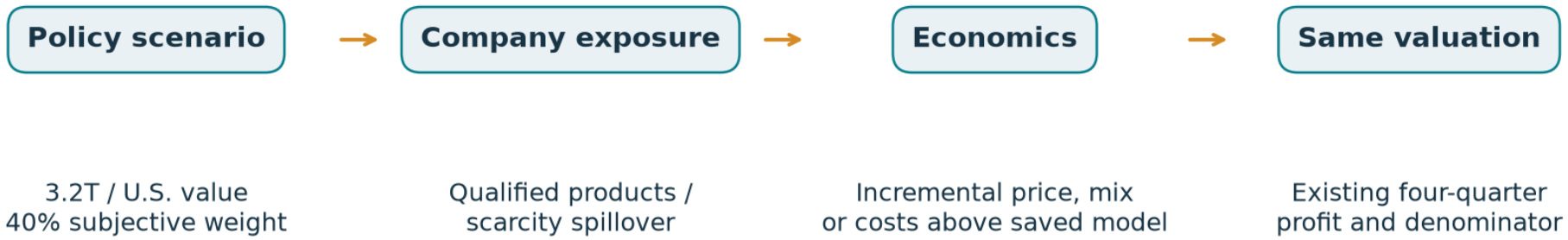


Figure 2. Analytical transmission of the attributed policy scenario. This is a model diagram, not an official regulatory diagram or customer award.

Assigned common policy-scenario probability: 40%. Subjective initial weight above the single-source 25-35% band because Serenity has the owner's highest weighting and attributes Washington meetings to Morgan Stanley. The exact future rule, timing and economic effect remain uncertain. Serenity and Morgan Stanley are one source chain, not two independent confirmations.

60% of data-center revenue is a bounded optical-content proxy, excluding the remaining copper/other exposure. Half is eligible and half repricable. Existing no-erosion pricing is the starting point. Keep the inherited 72% data-center gross-margin cap; additional cost/mix absorption prevents mechanically exceeding it.

Eligible fractions are bounded scenario assumptions, not origin certificates. U.S. corporate domicile and a U.S. listing do not prove manufacturing origin. If the final content definition excludes the modeled products, their eligible fraction goes to zero.

For component and integrated suppliers, the filtered revenue pool means 3.2T-linked sales plus incremental cross-generation pricing spillover from competition for common qualified capacity. It is not a claim that 800G/1.6T is directly regulated. Fractions are LOW-confidence exposure proxies, not verified geographic market shares.

Signal ledger field	Current treatment
Signal ID / date	FCC-OPTICS-20261001; Serenity/MS source chain
What is already included	All saved product growth, shortages, capacity, customer probabilities and contracts. The bridge adds only the explicitly calculated incremental economics.
Probability and timing	40% common policy scenario; price/cost intensity 0 before Q3 2027, 25% Q3 2027, 50% Q4 2027, 75% Q1 2028, 100% from Q2 2028.
Numerical contribution	Q4 2027 operating quarter: delta revenue 0.427500, delta consolidated OP 0.384750 (USD millions). Forward-four-quarter valuation OP changes 2.640050.
Dependence	The policy probability is shared across companies. These are correlated portfolio exposures, not independent confirmations. Existing customer-conversion probabilities are already inside source revenue.
Raise weight	Rule/proposal scope supports qualifying content and timing; company product/foundry evidence improves; customer sourcing or like-for-like price negotiations support the incremental premium.
Reduce weight	Policy is delayed beyond the horizon, the content definition excludes products, supply eases, fixed contracts prevent repricing, unit costs absorb the premium, or 3.2T conversion slips.
Zero-contribution boundary	No broad TAM uplift, no new factory capacity, no OCS benefit and no blanket-ban recovery.

Conditional inputs selected before calculating targets

Input	Value	Evidence status
Optical fraction of data-center pool	60.00%	LOW-confidence analyst assumption; inherited cap where labeled
Relevant qualifying exposure	50.00%	LOW-confidence analyst assumption; inherited cap where labeled
New / repricable contracts	50.00%	LOW-confidence analyst assumption; inherited cap where labeled
Conditional incremental price premium	5.00%	LOW-confidence analyst assumption; inherited cap where labeled
Incremental COGS / price revenue	5.00%	LOW-confidence analyst assumption; inherited cap where labeled
Incremental opex / incremental revenue	5.00%	LOW-confidence analyst assumption; inherited cap where labeled
Data-center gross-margin cap	72.00%	LOW-confidence analyst assumption; inherited cap where labeled

Incremental COGS and opex rates apply to incremental price revenue, not original product revenue or COGS.

Timing is an analyst sensitivity. A deliberately labeled early-sourcing proxy ahead of a 2028 3.2T ramp. Earlier quarters receive zero incremental policy revenue. Full intensity begins Q2 2028 and remains an uncertain continuation through the saved forecast only. These dates are not announced orders or a published regulatory timetable. The probability sensitivity below includes a no-policy control; a later start would reduce the near-term contribution.

Incremental revenue, cost and profit

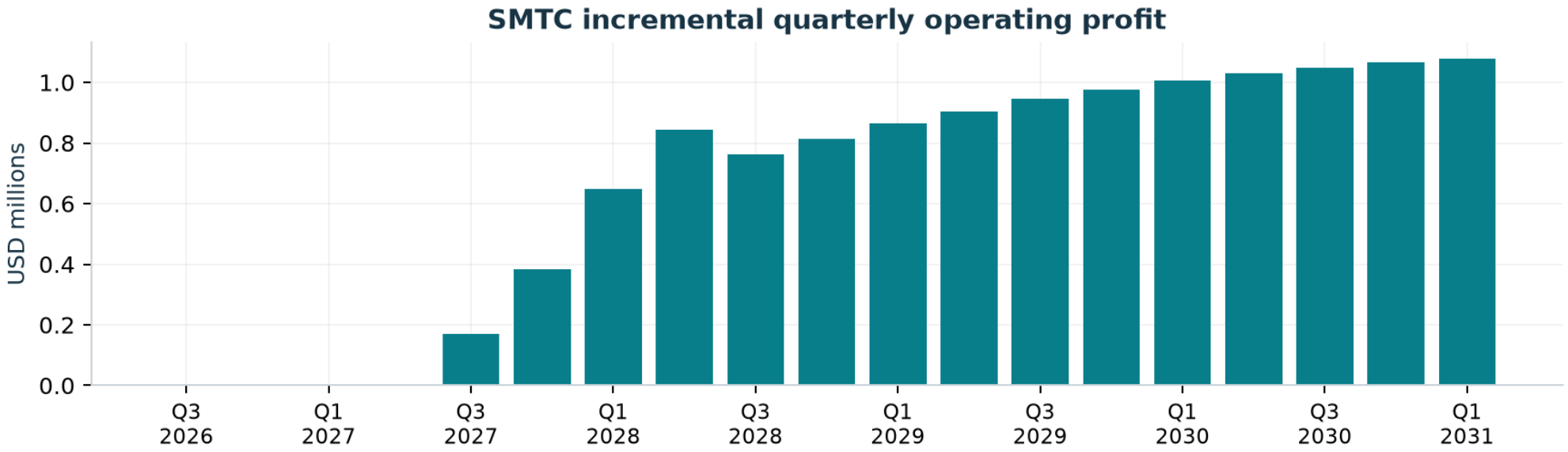


Figure 3. Incremental consolidated operating profit, USD millions. Bars show the probability-weighted incremental policy effect; the reference operating path is unchanged.

Quarter	Delta revenue	Delta COGS	Delta gross profit	Delta opex	Delta OP
Q3 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2027	0.1875	0.0094	0.1781	0.0094	0.1688
Q4 2027	0.4275	0.0214	0.4061	0.0214	0.3848
Q1 2028	0.7200	0.0360	0.6840	0.0360	0.6480
Q2 2028	1.0500	0.1540	0.8960	0.0525	0.8435
Q3 2028	1.1400	0.3192	0.8208	0.0570	0.7638
Q4 2028	1.2150	0.3402	0.8748	0.0608	0.8141
Q1 2029	1.2900	0.3612	0.9288	0.0645	0.8643
Q2 2029	1.3500	0.3780	0.9720	0.0675	0.9045
Q3 2029	1.4100	0.3948	1.0152	0.0705	0.9447
Q4 2029	1.4550	0.4074	1.0476	0.0728	0.9748
Q1 2030	1.5000	0.4200	1.0800	0.0750	1.0050
Q2 2030	1.5360	0.4301	1.1059	0.0768	1.0291
Q3 2030	1.5660	0.4385	1.1275	0.0783	1.0492
Q4 2030	1.5900	0.4452	1.1448	0.0795	1.0653
Q1 2031	1.6080	0.4502	1.1578	0.0804	1.0774

Active-window product bridge

Product / event	Pre-policy operating pool	Filtered exposure	Weighted delta revenue	Weighted delta OP
Data Center: FiberEdge, CopperEdge and photonics	285.0000	42.7500	0.4275	0.3848

Filtered exposure is a scenario allocation, not verified U.S.-origin revenue. All financial amounts use the company table units.

Business research and source evidence

The following sets out the company thesis, historical results, guidance, capacity, program probabilities, risks and original source images. Source dates remain visible. Operating assumptions below feed the complete current financial tables, including the explicit probability-weighted policy contribution.

1. Core thesis

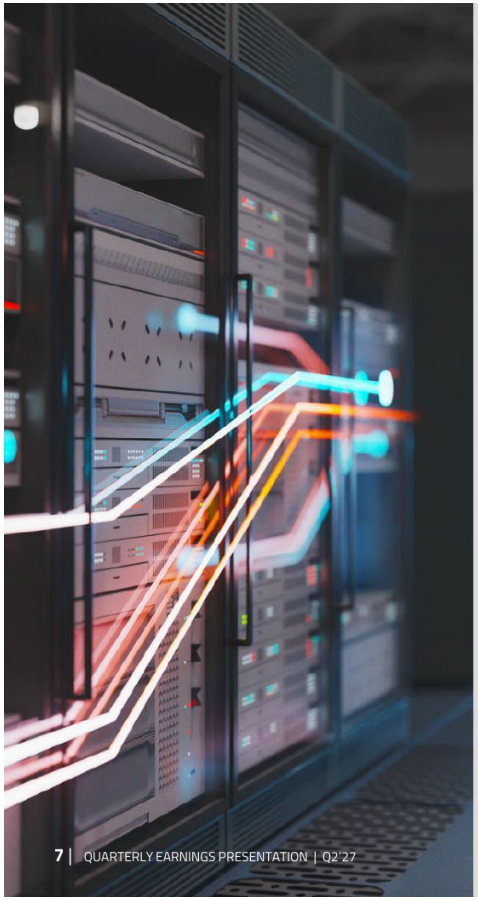
Semtech sells the analog and photonic layer that makes high-speed interconnect work: TIAs and drivers on the optical side, linear equalizers on the copper side, and now gain chips, CW lasers, SOAs and photodiodes underneath both.

Rule 9 test: if Semtech disappeared the architecture would still work, so this is a bottleneck rather than a chokepoint. MACOM, Broadcom and Marvell all build competing silicon. What has changed is the share: FiberEdge went from roughly 18% of 800G two years ago to **well over 50%**, the company is **designing to every module provider in its target markets, several on a sole-source basis**, and CopperEdge linear equalizers are described as the **de facto industry standard** with the lion's share of that market.

Supercycle: DB1 photonics and CPO. Signal-hierarchy position: Layer 3, critical component.

2. Latest earnings anchor (AH-15)

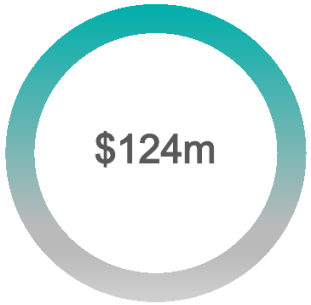
FQ2 FY2027, quarter ended 26 July 2026, reported 25 August. Exact issuer numbers below are separated from the rounded/calibrated model anchor.



Infrastructure End Market

Data Center, PON/FTTH, Wireless, Infrastructure Circuit Protection

Q2'27 NET SALES



Q/Q Change	+25%
Y/Y Change	+69%

- Record Q2'27 data center net sales of $100m, +39% Q/Q and +91% Y/Y; continued strength in 800G and early 1.6T ramp
- Strong demand for FiberEdge TIA and driver solutions; deepening engagement across all leading hyperscalers; now designed into every major module provider in target markets
- FiberEdge at 1.6T expected to exceed 50% market share by the end of FY'27
- CopperEdge up to 1.6T are ready for volume deployment; currently engaged in the design-in phase at all bandwidths up to 3.2T
- Photonics portfolio now spans gain chips, high-power CW lasers, SOAs and high-speed photodiodes for scale-up, scale-out and scale-across data center connectivity applications
- Positioned to grow content per transceiver from high single-digit to high double-digit dollars as the industry transitions from 800G to 3.2T
- Executed well on capacity expansion plan, securing equipment deliveries for this fiscal year and acquiring additional cleanroom space
- Targeting 45% Q/Q and 160% Y/Y data center net sales growth in Q3'27, with accelerating Y/Y growth and continued momentum through FY'28

SEMTECH

Source exhibit C4. Semtech FQ2 FY2027 presentation, August 25, 2026, PDF page 7. Reported infrastructure / data-center results and management commentary.

Metric	Value	Note
Revenue	**341.9**	issuer-reported; record, above the high end of outlook, +17% sequential, +33% YoY. **Tenth consecutive quarter of growth**
Data Center	**100.0**	record, **+39% sequential, +91% YoY**
Infrastructure	124.0	+25% sequential, +69% YoY
LoRa-enabled	**58.0**	all-time record, +31% sequential, +58% YoY
IoT systems and connectivity	98.0	+11% sequential and YoY
High-end consumer	39.0	+2% sequential, −5% YoY
Adjusted gross margin	**54.5%**	+150bp sequential. Semiconductor products **62.8%**, +210bp. **Ex-module 59.7%**
Adjusted net opex	102.8	issuer-reported; below the low end of guidance
Adjusted operating income	**83.6**	issuer-reported; **24.4% margin**
Adjusted EBITDA	91.0	26.6% margin
Adjusted diluted EPS	**0.71**	+39% sequential, **+73% YoY**, more than twice the rate of revenue growth
Operating cash flow / FCF	68.9 / 61.4	issuer-reported; net capex 7.5
Cash / debt principal	204.0 / 503.0	net leverage 1.1

Q3’27 Outlook*

(in millions, except per share data)						
Net sales	$	410.0		+/-	$	5.0
Non-GAAP Financial Measures						
Adjusted gross margin*		58.3	%	+/-		100 bps
*Adjusted gross margin, excluding held-for-sale**		63.9	%	+/-		100 bps
Adjusted operating expenses, net*	$	112.0		+/-	$	3.0
Adjusted operating income*	$	127.0		+/-	$	4.0
Adjusted operating margin*		31.0	%	+/-		60 bps
Adjusted interest and other expense, net*	$	0.5				
Adjusted normalized tax rate*		18	%			
Adjusted diluted earnings per share*	$	1.05		+/-	$	0.03
Adjusted EBITDA*	$	134.3		+/-	$	4.0
Adjusted EBITDA margin*		32.8	%	+/-		60 bps
Non-GAAP diluted share count*		99.0				

*See Non-GAAP Financial Measures above

SEMTECH

Source exhibit C5. Semtech FQ2 FY2027 presentation, August 25, 2026, PDF page 16. Management FQ3 outlook, not reported future results.

FQ3 FY2027 guidance, and the model against it

	Guided	Model
Revenue	410 ± 5	**410.0**
Blended adjusted gross margin	58.3% ± 100bp	**58.21%**
Adjusted net opex	112 ± 3	**112.0**
Adjusted operating margin	31.0%	**30.89%**
Adjusted diluted EPS	1.05 ± 0.03	**1.04**
Gross margin excluding modules	**63.9%**	**63.86%**

Data centre is guided **+45% sequentially, or +160% year over year**, and LoRa **+15% sequentially, +65% year over year**. The near-term reference reproduces the disclosed guidance lines, including the ex-module cross-check; the new policy inputs begin later.

Exact issuer results and the saved model anchor

The actual period is **FQ2 FY2027 ended July 26, 2026, reported August 25**. The saved model labels it Q3 2026; this inherited mapping is a financial-model slot, not an exact July-September reported quarter. The next slot aligns to FQ3 FY2027 guidance. Future product allocations remain modeled, even when the consolidated total is anchored to guidance.

Metric, USD M unless stated	Issuer FQ2 FY2027	Saved Q3 2026 model anchor	Interpretation
Revenue	341.9	342.0000	Rounded / calibrated model total
Adjusted operating profit	83.6	83.9992	Saved adjusted-OP anchor; not exact reported OP
Adjusted opex	102.8	102.3948	Saved calibrated cost anchor
Held-for-sale revenue / adjusted GP	48.1 / 10.9	40.0 / 6.0	Explicit issuer disclosure versus legacy module allocation
AirLink / retained IoT revenue	Not separately assigned here	58.0	Legacy model allocation
DC-only gross margin	Not disclosed	64.8%	Nonunique model residual
Combined semiconductor GM	62.8%	Calibration reference	Broader than DC alone
Signal Integrity revenue / GP / GM	126.2 / 82.4 / 65.3%	No direct one-to-one vector mapping	Reported segment; not pure DC

The capacity evidence update preserves the saved product schedule, costs, shares and ceiling mechanics. The source-label corrections above prevent rounded totals and allocation residuals from being presented as exact issuer disclosures. Source: Semtech FQ2 FY2027 issuer presentation, PDF pages 17-18 and 21-23.

Reconciliation of GAAP to Non-GAAP Results

	Q2'27					
	Signal Integrity	**Analog Mixed Signal and Wireless**	**Total Semiconductor Products**	**IoT Systems and Connectivity**	**Unallocated[1]**	**Total**
Net sales	$ 126.2	$ 117.4	$ 243.5	$ 98.3	$ —	$ 341.9
Segment cost of sales	43.8	46.8	90.6	64.9	2.6	158.1
Segment gross profit	$ 82.4	$ 70.5	$ 153.0	$ 33.4	$ (2.6)	$ 183.8
Segment gross margin	65.3%	60.1%	62.8%	34.0%	NM[2]	
Gross margin (GAAP)						**53.8 %**
Share-based compensation						0.2 %
Amortization of acquired technology						0.5 %
Adjusted gross margin (Non-GAAP)						**54.5 %**
Business held for sale[3]						5.2 %
Adjusted gross margin, excluding business held for sale (Non-GAAP)						**59.7 %**

[1] Unallocated includes share-based compensation and amortization of acquired technology
[2] Not meaningful
[3] Adjustment for net sales of $48.1 million and adjusted gross profit of $10.9 million related to business held for sale as of Q2'27

Dollar amounts in millions and may not add precisely due to rounding

SEMTECH

Source exhibit C3. Semtech FQ2 FY2027 presentation, August 25, 2026, PDF page 18. Issuer segment reconciliation and held-for-sale figures; DC-only gross margin is not disclosed. Source URL appears in the source register.

3. Seven-vector decomposition (AH-11)

Vector	Anchor revenue	Anchor GM	Driver	AH-7
Data Center	100.0	64.8%	FiberEdge TIA and driver, CopperEdge linear equalizers, HieFo photonics. 1.6T surpasses 50% of data-centre revenue in FQ3	Anchor HIGH, path LOW
Infrastructure other	24.0	57.0%	PON, 5G, satellite	LOW
LoRa-enabled	58.0	70.0%	LoRaWAN, LoRa Plus at 20-25% of mix, Amazon Sidewalk at high single digits	Anchor HIGH, path LOW
AirLink / IoT connectivity	58.0	46.8%	5G RedCap routers, network slicing, software platform	LOW
Cellular modules	40.0	15.0%	**Held for sale.** Legacy model allocation; zero from calendar Q2 2027	LOW; model allocation
Industrial other	23.0	57.0%	Broad industrial analog	LOW
High-end consumer	39.0	54.0%	TVS circuit protection, SurgeSwitch, PerSe capacitive sensing	LOW

The saved anchor allocations are model assumptions, not uniquely solved issuer disclosures. The 40.0 module and 58.0 AirLink revenue allocations, 15.0% module margin, 46.8% AirLink margin and **64.8% data-center margin** remain unchanged for this capacity-only update. The DC margin is a nonunique residual conditional on the other product assumptions. The issuer actually disclosed 48.1 of held-for-sale revenue and 10.9 of adjusted gross profit; it did not disclose DC-only gross profit. These differences are source-label caveats, not a forecast reset. All amounts are USD millions.

The module divestiture is a margin event, not a revenue event

A definitive agreement is signed, the business sits as held for sale, close is expected in FQ4 FY2027, and the transaction is **expected to be EPS neutral on a non-GAAP basis**. Management gives the bridge twice: 54.5% against 59.7% ex-module at the anchor, and 58.3% against 63.9% in the guide. That is a **520 to 560 basis point drag**. The issuer disclosed 48.1 of held-for-sale revenue and 10.9 of adjusted GP in FQ2 FY2027. The saved future allocation carries 43, then a part quarter of 21, then zero; these are retained model assumptions rather than a uniquely implied or disclosed future module schedule.

Mark Lin's framing: a good starting point post-divestiture is that **64% gross margin**.

Above the 64% starting point, disclosed under AH-15 clause 5

The inherited FY2028 blended gross margin was **66.40%**, before the incremental policy pricing contribution, above management's stated 64%. The permitted evidence is management-stated and specific: 64% is described as a *starting point* rather than a level; mix is named as the driver with 1.6T inflecting and LoRa compounding; there is **no price erosion on 800G**; **availability matters more to customers than pricing** and cost increases are being passed through; and CW lasers are described as carrying a data-centre gross margin accretive to the corporate average.

Data-centre gross margin is capped at 72.0% in this model. An unbounded mix effect would otherwise carry the vector above anything a merchant analog supplier sustains.

4. Rolling forward-4Q ceiling (AH-16)

Ceiling = [OP(Q) + OP(Q+1) + OP(Q+2) + OP(Q+3)] × 20 ÷ diluted shares

6. Key risks (each scored)

#	Risk	Detail and kill switch
1	**Competitive socket**	MACOM, Broadcom and Marvell all build competing TIAs and drivers. Share above 50% is a position, not a barrier, and it was won inside three years, which means it can be lost inside three years. **Kill switch:** FiberEdge 1.6T share failing to exceed 50% by fiscal year end.
2	**Capacity may not be enough**	Management states directly that secured capacity **may not be enough for FY2028, especially the second half**, and is using prepayments and capex with front-end and back-end partners to secure more. Room is described as 50 to 100% additional. Revenue is capped by supply, not demand. **Kill switch:** any quarter where revenue is constrained by capacity rather than orders.
3	**Photonics execution**	HieFo has been owned five to six months. CW lasers and SOAs are sampling within a couple of months and **revenue does not begin until H1 FY2028**. Photodiode arrays are a design team, not a product. Fab capacity must rise 3 to 4x by year end. **Kill switch:** CW laser transceiver revenue slipping beyond H1 FY2028.
4	**Convertible dilution**	The share count rises from 98.5M to a modelled 107.4M across the ladder as convertibles enter. Every rung is divided by a larger number than the last. **Kill switch:** diluted share count exceeding 110M.

Not scored: the cellular module divestiture. It removes roughly 43 of quarterly revenue and adds more than 500 basis points of structural gross margin, and management expects it to be EPS neutral on a non-GAAP basis.

7. Cross-book

Name	Read-through
AAOI	Buys Semtech TIAs and drivers. Semtech confirming exceptionally strong FiberEdge demand and earlier-than-expected 1.6T qualification is a direct read-through to AAOI's own 1.6T ramp, and both describe the same customer urgency.
LITE, COHR	Module makers and, in the CW laser layer, prospective competitors. Semtech's HieFo lasers target the same external-light-source socket, though revenue is a year out.
SIVE.ST	The purest CW laser position. Semtech entering high-power CW lasers with five or six module manufacturers evaluating is the first credible merchant challenge to that socket, and the timing is H1 FY2028.
MRVL	Competes in optical DSP and retimed architectures. Semtech's linear and LPO positioning is the counter-architecture: linear equalizers displace retimers where link margin allows.
3363.TWO FOCI, 6451.TW ShunSin	The CPO packaging layer Semtech's NPO and MPO engagement feeds into.
MTSI	The closest listed comparable in high-speed analog for optical, and the direct share counterparty in TIAs and drivers.

8. Open rulings

#	Ruling required	Impact
1	**The 20× multiple** is carried as the sanctioned input for this name and is not re-derived here	Current Q4 2027 sensitivity: 18x gives USD184.60; 25x gives USD256.39. Selected method remains 20x
2	**Inherited FY2028 gross margin of 66.4% against management's stated 64% starting point.** Disclosed under AH-15 clause 5 on mix, pricing and sole-source evidence	A flat 64% assumption would lower profit; the current model tables provide the revised margin path
3	**The seven-vector split is modeled.** Segment totals, DC and LoRa sales, and the held-for-sale business revenue and GP are disclosed; DC-only GP is not	Saved allocations are retained for this capacity update; the 64.8% DC margin is a nonunique model residual
4	**Data-centre gross margin capped at 72.0%** by judgement rather than by disclosure	The cap limits this current model increment; removing it is not part of the selected case
5	**Future module revenue of 43 and the modeled 10-15% margins** are retained assumptions; issuer FQ2 held-for-sale revenue / GP were 48.1 / 10.9	Historical source-label correction does not reset the frozen forecast allocations

9. Re-rate triggers

#	Trigger	Test	If passed	If failed
1	**Investor event, 15 October 2026**	The multi-year financial framework and long-term targets, plus data-centre TAM and share by application	Anchor the FY2029-31 path to management's own framework	Model stands on its own construction
2	**FQ3 FY2027 results**	Revenue against the guided 410 ± 5 and data centre against the implied 145	Retag FQ3 from GUIDED to ACTUAL; anchor the ramp	Cut the data-centre path
3	FiberEdge 1.6T share	Above 50% by fiscal year end as management states	Core Thesis evidence confirmed	Risk 1 binds
4	Divestiture close	Completion in FQ4 FY2027 and the reported ex-module margin structure	Margin step confirmed in reported numbers	Reprice the margin bridge
5	CW laser revenue	First transceiver revenue in H1 FY2028	**Rule 16b escalation pre-committed: photonics vector 25% → 60%** on first disclosed revenue	Risk 3 binds
6	Capacity	Any disclosed prepayment or capacity agreement with front-end or back-end partners	Removes the FY2028 second-half supply cap	Risk 2 binds
7	Counter-trigger	A named competitor winning a sole-source socket Semtech currently holds, or 800G price erosion appearing	-	Immediate Core Thesis and margin review

Business analysis originally built 2026-08-26 from Semtech's FQ2 FY2027 results and earnings call (2026-08-25), with source-label corrections and capacity evidence reviewed 2026-10-03. Anchor product margins and the saved module / AirLink split are model assumptions. Historical evidence retains its source dates. The retained financial tables reconcile to SMTC_Q_Model in the master workbook.

Complete current operating model

Status and convention: the Q3 2026 slot is a rounded/calibrated anchor for FQ2 FY2027 ended July 26; it is not exact reported calendar-quarter data. Q4 2026 is guided at the consolidated level with modeled product allocations; Q1 2027 onward is forecast. All 19 saved rows, seven product paths, costs, margins, shares and 16 complete ceiling windows are retained. The capacity ledger adds zero incremental economics; the existing policy ledger remains included once.

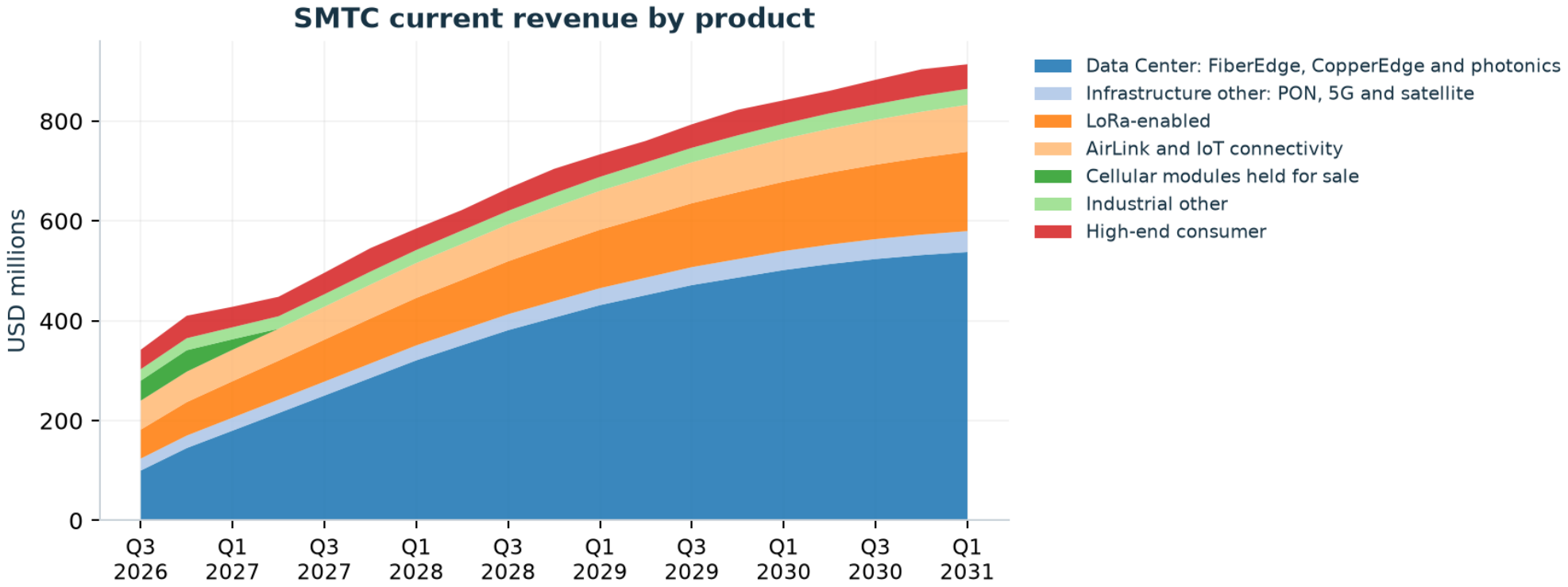


Figure 4. Current quarterly revenue by product, USD millions. All seven Semtech product vectors are shown.

Quarter	Revenue	Gross profit	COGS
Q3 2026	342.000	186.394	155.606
Q4 2026	410.000	238.650	171.350
Q1 2027	428.000	265.693	162.307
Q2 2027	448.000	293.622	154.378
Q3 2027	496.188	328.142	168.045
Q4 2027	545.428	363.335	182.092
Q1 2028	584.720	392.664	192.056
Q2 2028	622.050	420.216	201.834
Q3 2028	665.140	450.775	214.365
Q4 2028	704.215	478.145	226.070
Q1 2029	733.290	499.671	233.619
Q2 2029	760.350	519.342	241.008
Q3 2029	793.410	542.269	251.141
Q4 2029	822.455	562.312	260.143
Q1 2030	841.500	576.351	265.149
Q2 2030	860.536	590.090	270.446
Q3 2030	882.566	605.108	277.458
Q4 2030	903.590	619.417	284.173
Q1 2031	913.608	626.983	286.625

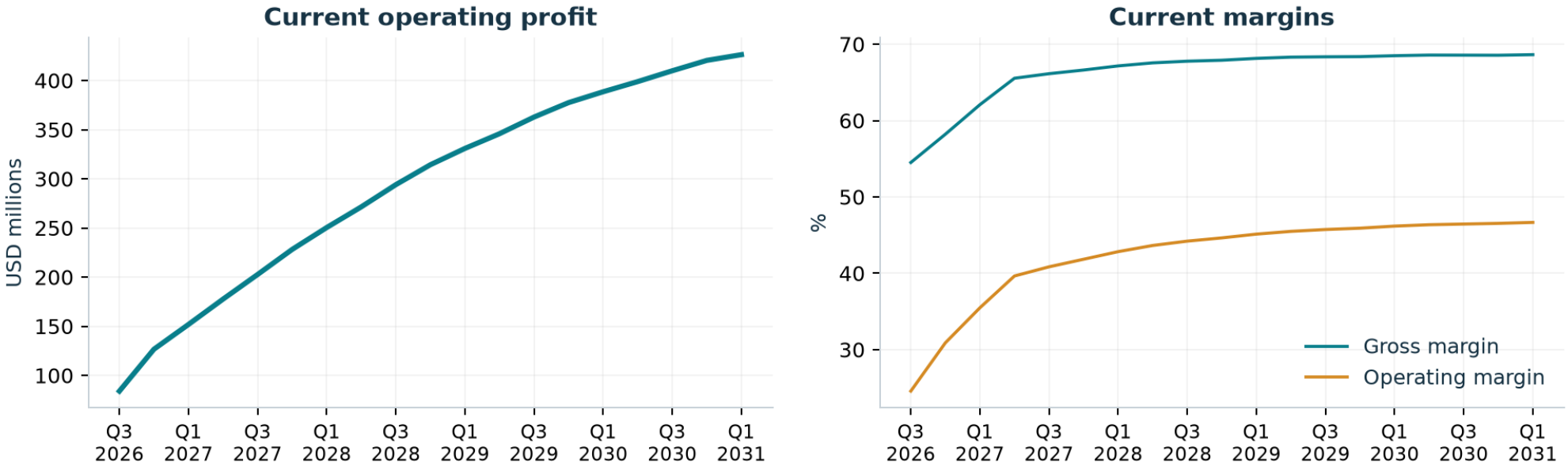


Figure 5. Current operating profit and margins. Operating profit is consolidated adjusted operating profit in USD millions, using the retained Semtech valuation basis.

Quarter	Revenue	Gross margin	Opex	Operating profit	OP margin	Valuation OP
Q3 2026	342.000	54.50%	102.395	83.999	24.56%	83.999
Q4 2026	410.000	58.21%	112.012	126.638	30.89%	126.638
Q1 2027	428.000	62.08%	113.848	151.845	35.48%	151.845
Q2 2027	448.000	65.54%	116.032	177.590	39.64%	177.590
Q3 2027	496.188	66.13%	125.497	202.645	40.84%	202.645
Q4 2027	545.428	66.61%	135.181	228.154	41.83%	228.154
Q1 2028	584.720	67.15%	142.240	250.424	42.83%	250.424
Q2 2028	622.050	67.55%	148.782	271.434	43.64%	271.434
Q3 2028	665.140	67.77%	156.761	294.014	44.20%	294.014
Q4 2028	704.215	67.90%	163.860	314.285	44.63%	314.285
Q1 2029	733.290	68.14%	168.791	330.880	45.12%	330.880
Q2 2029	760.350	68.30%	173.499	345.843	45.48%	345.843
Q3 2029	793.410	68.35%	179.459	362.811	45.73%	362.811
Q4 2029	822.455	68.37%	184.798	377.514	45.90%	377.514
Q1 2030	841.500	68.49%	187.815	388.536	46.17%	388.536
Q2 2030	860.536	68.57%	191.204	398.886	46.35%	398.886
Q3 2030	882.566	68.56%	195.220	409.888	46.44%	409.888
Q4 2030	903.590	68.55%	198.970	420.446	46.53%	420.446
Q1 2031	913.608	68.63%	200.720	426.262	46.66%	426.262

Every revenue vector and product margin

Product GP = product revenue x the displayed modeled margin. Product allocations and future margins remain LOW-confidence model assumptions unless specifically identified as historical or guided.

Data Center: FiberEdge, CopperEdge and photonics

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	100.000	64.80%	64.800
Q4 2026	145.000	70.00%	101.500
Q1 2027	180.000	70.40%	126.720
Q2 2027	215.000	70.80%	152.220
Q3 2027	250.188	71.22%	178.178
Q4 2027	285.428	71.54%	204.181
Q1 2028	320.720	71.75%	230.124
Q2 2028	351.050	71.94%	252.546
Q3 2028	381.140	72.00%	274.421
Q4 2028	406.215	72.00%	292.475
Q1 2029	431.290	72.00%	310.529
Q2 2029	451.350	72.00%	324.972
Q3 2029	471.410	72.00%	339.415
Q4 2029	486.455	72.00%	350.248
Q1 2030	501.500	72.00%	361.080
Q2 2030	513.536	72.00%	369.746
Q3 2030	523.566	72.00%	376.968
Q4 2030	531.590	72.00%	382.745
Q1 2031	537.608	72.00%	387.078

Infrastructure other: PON, 5G and satellite

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	24.000	57.00%	13.680
Q4 2026	25.000	59.00%	14.750
Q1 2027	26.000	59.10%	15.366
Q2 2027	27.000	59.20%	15.984
Q3 2027	28.000	59.30%	16.604
Q4 2027	29.000	59.40%	17.226
Q1 2028	30.000	59.50%	17.850
Q2 2028	31.000	59.60%	18.476
Q3 2028	32.000	59.70%	19.104
Q4 2028	33.000	59.80%	19.734
Q1 2029	34.000	59.90%	20.366
Q2 2029	35.000	60.00%	21.000
Q3 2029	36.000	60.00%	21.600
Q4 2029	37.000	60.10%	22.237
Q1 2030	38.000	60.10%	22.838
Q2 2030	39.000	60.20%	23.478
Q3 2030	40.000	60.20%	24.080
Q4 2030	41.000	60.30%	24.723
Q1 2031	42.000	60.30%	25.326

LoRa-enabled

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	58.000	70.00%	40.600
Q4 2026	67.000	72.00%	48.240
Q1 2027	73.000	72.10%	52.633
Q2 2027	78.000	72.20%	56.316
Q3 2027	84.000	72.30%	60.732
Q4 2027	90.000	72.40%	65.160
Q1 2028	95.000	72.50%	68.875
Q2 2028	100.000	72.60%	72.600
Q3 2028	106.000	72.70%	77.062
Q4 2028	112.000	72.80%	81.536
Q1 2029	117.000	72.90%	85.293
Q2 2029	122.000	73.00%	89.060
Q3 2029	128.000	73.00%	93.440
Q4 2029	134.000	73.00%	97.820
Q1 2030	139.000	73.00%	101.470
Q2 2030	144.000	73.00%	105.120
Q3 2030	149.000	73.00%	108.770
Q4 2030	154.000	73.00%	112.420
Q1 2031	159.000	73.00%	116.070

AirLink and IoT connectivity

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	58.000	46.80%	27.144
Q4 2026	61.000	50.00%	30.500
Q1 2027	63.000	50.30%	31.689
Q2 2027	64.000	50.60%	32.384
Q3 2027	66.000	50.90%	33.594
Q4 2027	68.000	51.20%	34.816
Q1 2028	70.000	51.50%	36.050
Q2 2028	72.000	51.80%	37.296
Q3 2028	74.000	52.10%	38.554
Q4 2028	76.000	52.40%	39.824
Q1 2029	78.000	52.70%	41.106
Q2 2029	80.000	53.00%	42.400
Q3 2029	82.000	53.20%	43.624
Q4 2029	84.000	53.40%	44.856
Q1 2030	86.000	53.60%	46.096
Q2 2030	88.000	53.80%	47.344
Q3 2030	90.000	54.00%	48.600
Q4 2030	92.000	54.20%	49.864
Q1 2031	94.000	54.40%	51.136

Cellular modules held for sale

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	40.000	15.00%	6.000
Q4 2026	43.000	10.00%	4.300
Q1 2027	21.000	10.00%	2.100
Q2 2027	0.000	0.00%	0.000
Q3 2027	0.000	0.00%	0.000
Q4 2027	0.000	0.00%	0.000
Q1 2028	0.000	0.00%	0.000
Q2 2028	0.000	0.00%	0.000
Q3 2028	0.000	0.00%	0.000
Q4 2028	0.000	0.00%	0.000
Q1 2029	0.000	0.00%	0.000
Q2 2029	0.000	0.00%	0.000
Q3 2029	0.000	0.00%	0.000
Q4 2029	0.000	0.00%	0.000
Q1 2030	0.000	0.00%	0.000
Q2 2030	0.000	0.00%	0.000
Q3 2030	0.000	0.00%	0.000
Q4 2030	0.000	0.00%	0.000
Q1 2031	0.000	0.00%	0.000

Industrial other

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	23.000	57.00%	13.110
Q4 2026	24.000	59.00%	14.160
Q1 2027	24.000	59.10%	14.184
Q2 2027	25.000	59.20%	14.800
Q3 2027	25.000	59.30%	14.825
Q4 2027	26.000	59.40%	15.444
Q1 2028	26.000	59.50%	15.470
Q2 2028	27.000	59.60%	16.092
Q3 2028	27.000	59.70%	16.119
Q4 2028	28.000	59.80%	16.744
Q1 2029	28.000	59.90%	16.772
Q2 2029	29.000	60.00%	17.400
Q3 2029	29.000	60.00%	17.400
Q4 2029	30.000	60.10%	18.030
Q1 2030	30.000	60.10%	18.030
Q2 2030	31.000	60.20%	18.662
Q3 2030	31.000	60.20%	18.662
Q4 2030	32.000	60.30%	19.296
Q1 2031	32.000	60.30%	19.296

High-end consumer

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	39.000	54.00%	21.060
Q4 2026	45.000	56.00%	25.200
Q1 2027	41.000	56.10%	23.001
Q2 2027	39.000	56.20%	21.918
Q3 2027	43.000	56.30%	24.209
Q4 2027	47.000	56.40%	26.508
Q1 2028	43.000	56.50%	24.295
Q2 2028	41.000	56.60%	23.206
Q3 2028	45.000	56.70%	25.515
Q4 2028	49.000	56.80%	27.832
Q1 2029	45.000	56.90%	25.605
Q2 2029	43.000	57.00%	24.510
Q3 2029	47.000	57.00%	26.790
Q4 2029	51.000	57.10%	29.121
Q1 2030	47.000	57.10%	26.837
Q2 2030	45.000	57.20%	25.740
Q3 2030	49.000	57.20%	28.028
Q4 2030	53.000	57.30%	30.369
Q1 2031	49.000	57.30%	28.077

Full ceiling derivation

Valuation OP equals consolidated OP except where the saved model allocates economics to parent shareholders. The current model adds ownership x incremental OP to the saved parent-economic OP. Existing parent-only expenses and senior claims remain deducted as before. For every ceiling, sum that quarter and its next three quarters, multiply by the sanctioned multiple and currency/unit factor, and divide by the saved quarter-specific share denominator. No stock-price-dependent upside or ranking appears in this artifact.

Ceiling quarter / window	F4Q valuation OP	Multiple	Shares M	FX / unit factor	Ceiling
Q3 2026 to Q2 2027	540.072200	20x	98.500000	1.000000	109.659330
Q4 2026 to Q3 2027	658.717750	20x	99.000000	1.000000	133.074293
Q1 2027 to Q4 2027	760.233500	20x	99.500000	1.000000	152.810754
Q2 2027 to Q1 2028	858.812500	20x	100.200000	1.000000	171.419661
Q3 2027 to Q2 2028	952.656500	20x	101.000000	1.000000	188.644851
Q4 2027 to Q3 2028	1,044.025550	20x	101.800000	1.000000	205.113075
Q1 2028 to Q4 2028	1,130.156850	20x	102.600000	1.000000	220.303480
Q2 2028 to Q1 2029	1,210.613150	20x	103.400000	1.000000	234.161151
Q3 2028 to Q2 2029	1,285.022150	20x	104.100000	1.000000	246.882257
Q4 2028 to Q3 2029	1,353.819050	20x	104.800000	1.000000	258.362414
Q1 2029 to Q4 2029	1,417.047850	20x	105.400000	1.000000	268.889535
Q2 2029 to Q1 2030	1,474.703550	20x	106.000000	1.000000	278.245953
Q3 2029 to Q2 2030	1,527.746170	20x	106.500000	1.000000	286.900689
Q4 2029 to Q3 2030	1,574.823190	20x	107.000000	1.000000	294.359475
Q1 2030 to Q4 2030	1,617.755640	20x	107.400000	1.000000	301.258034
Q2 2030 to Q1 2031	1,655.482000	20x	107.700000	1.000000	307.424698

Q4 2027 worked example: (228.153750 + 250.424000 + 271.434000 + 294.013800) x 20 x 1 / 101.800000 = **205.113075 USD/share**.

Annual cross-check

Calendar-year sums below include only the quarters in the saved model. Partial years are labeled. These are not additional revenue, a horizon roll, or new management guidance.

Calendar year	Coverage	Revenue	Gross profit	Operating profit	OP margin
2026	2 quarters	752.000	425.044	210.637	28.01%
2027	4 quarters	1,917.615	1,250.792	760.233	39.64%
2028	4 quarters	2,576.125	1,741.800	1,130.157	43.87%
2029	4 quarters	3,109.505	2,123.594	1,417.048	45.57%
2030	4 quarters	3,488.192	2,390.965	1,617.756	46.38%
2031	1 quarters	913.608	626.983	426.262	46.66%

Probability and dilution sensitivity

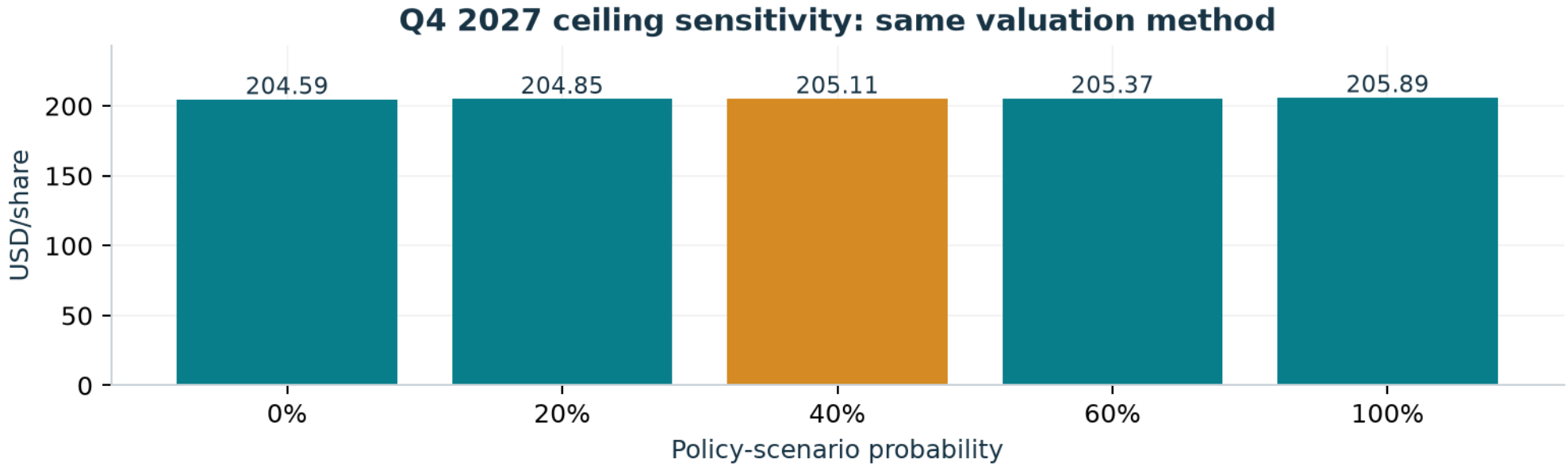


Figure 6. Q4 2027 ceiling under alternative policy probabilities. Gold is the selected 40% weight. All company valuation methods, operating reference inputs and shares remain unchanged.

Policy probability	Q4 2027 ceiling	Change from zero-policy sensitivity
0%	204.5944	0.000%
20%	204.8537	0.127%
40%	205.1131	0.254%
60%	205.3724	0.380%
100%	205.8911	0.634%

Additional diluted shares	Q4 2027 Ceiling	Interpretation
0%	205.1131	Same operating case; denominator sensitivity only
5%	195.3458	Same operating case; denominator sensitivity only
10%	186.4664	Same operating case; denominator sensitivity only

The 40% weight is a judgment about the incremental policy economics and timing, not a statistical certainty. Because the nine companies share the same policy event, these increments should not be treated as nine independent sources of portfolio diversification. The next 12-16 months have the highest forecasting relevance; later continuation assumptions are progressively less reliable.

Validation and next evidence

This current artifact and the master model use the same selected 40% policy case. Existing company valuation methods, quarter conventions, ownership and dilution schedules are retained. Every existing portfolio Excel Base is protected; the analytical score above is assessed separately.

The incremental ledger exposes revenue, production costs, gross profit and operating expense. A zero-policy sensitivity removes only that ledger's contribution. It is a diagnostic control, not a competing active forecast. Reported facts and dated guidance retain their original dates; new pricing, exposure, costs and timing are labeled analyst assumptions.

The most useful next evidence is the proposed content definition, product-specific fabrication and assembly routes, contract repricing rights, delivered 3.2T qualification, accepted component premiums and unit-cost/yield data. Those observations can raise or lower the inputs and their probability; official confirmation is not a prerequisite for modeling this credible breadcrumb.

Current model: **SMTC_Q_Model** in **v 3_2_master_portfolio.xlsx**. The full quarterly model includes all original product vectors and the approved incremental operating bridge. Approval snapshots and change history are retained externally rather than as competing current forecasts.

Source register for the October 3 evidence update

ID	Source / date / location	Evidence status
C1	P Equity Research / tuolaji, Expert Call: Tower Semiconductor, October 2, 2026, supplied 23-page PDF, pages 17-18	Anonymous expert / AI-translated channel evidence; not issuer disclosure
C2	Frozen Semtech capacity calibration, October 3, 2026	Analyst bridge from attributed inputs; not separate revenue forecast

- q 2_fy 2027_10q.pdf. Retrieved October 3, 2026. Issuer-hosted filing for July 26, 2026; PDF pages 4, 6, 9, 13, 22-27 and 47-51.
- q 2_fy 2027_presentation.pdf. Retrieved October 3, 2026. Issuer presentation dated August 25; PDF pages 7, 16-18, 21-23 and 27-28 are relevant.
- q 2_fy 2027_call_mirror.html. Retrieved October 3, 2026. August 25 call, third-party transcript copy; not independently transcribed audio.
- issuer_events.html. Retrieved October 3, 2026.
- issuer_webcast.html. Retrieved October 3, 2026. Issuer-linked webcast; retrieved landing page is registration-gated.
- Semtech September 21 quad / octal optical interconnect release. Primary product evidence; four dies per module is not universal.